# Genome-wide association of sub foveal outer segment thickness

Alvin E. Lirio[1]*, Callum Hunt[1], Ha-Jun Yoon[1], Lilian Zevlaris[1], Kayesha Coley[2,3], Chiara Batini[2,3], Mervyn G Thomas[1,3,4]

[1]Ulverscroft Eye Unit, School of Psychology and Vision Sciences, University of Leicester, Leicester, United Kingdom; [2]Department of Population Health Sciences, University of Leicester, Leicester, United Kingdom; [3]University Hospitals of Leicester NHS Trust, Leicester, United Kingdom; [4]Department of Ophthalmology, University Hospitals of Leicester NHS Trust, Leicester, United Kingdom

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## PURPOSE

**Why does outer segment (OS) thickness matter?**

The sub foveal OS is the light-capturing compartment of foveal cones. **OS thinning** is an early, measurable marker of photoreceptor damage in AMD and inherited retinal dystrophies, directly correlating with visual acuity loss[1].

The genetic architecture of sub foveal OS thickness is largely unknown. UK Biobank OCT imaging (>67,000 participants) provides an unprecedented resource for population-scale genetic analysis[2].

## METHODS

**Defining phenotype**

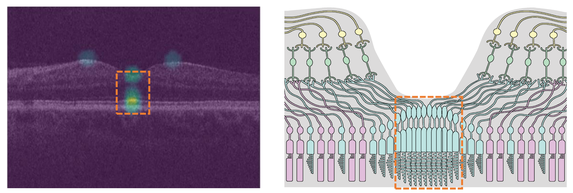


**Figure 1**: The sub foveal OS thickness was derived from the OCT imaging performed at the UK Biobank imaging assessment. The mean of left and right eye measurements was computed for individuals with valid measurements in at least one eye.

**Genome-wide association testing**

54,042 participants

**Gene prioritisation**
ANNOVAR |
Genotype–Phenotype Map[3]

**Look-up Analysis**
N=31,135 European ancestry |
15 OS loci | Currant, et al 2023

## Genome-wide-association study

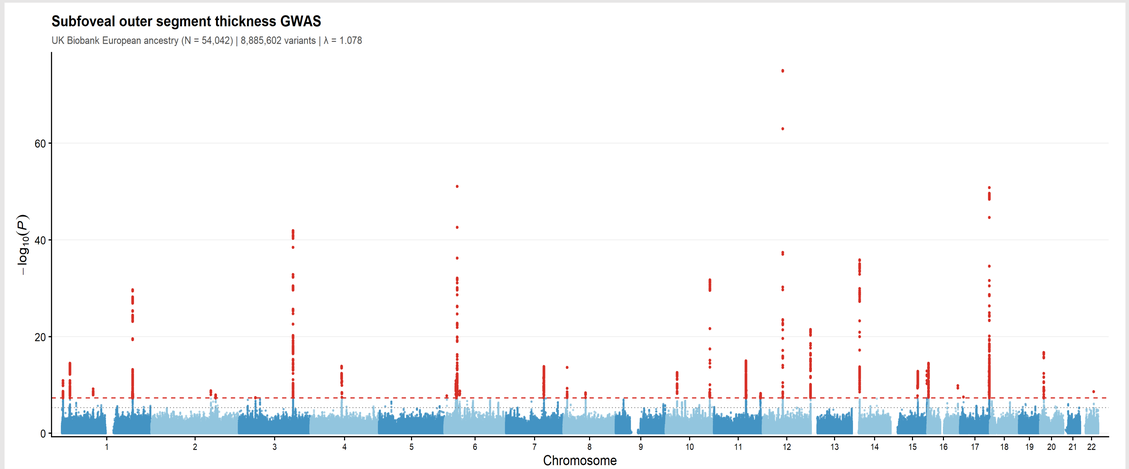



**Figure 2**: Manhattan Plot. The genome-wide significance threshold is set at $p<5\times10^{-8}$ (red dash horizontal line). Associations are plotted as $-\log_{10}(p\text{-value})$ on the y-axis, with chromosome position on the x-axis.

## Gene prioritisation: Pleiotropy & Functional validation

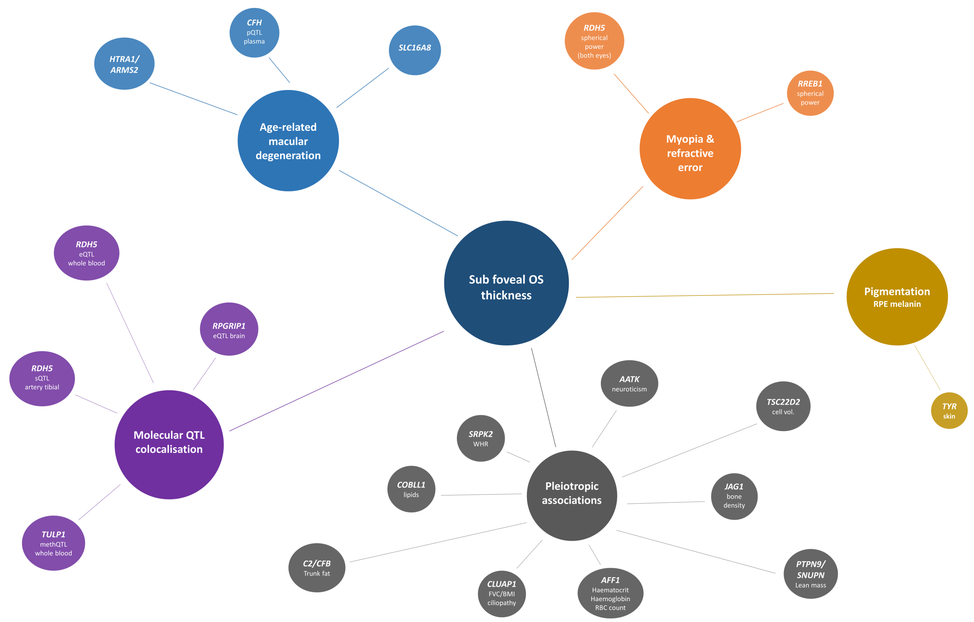



**Figure 3**: Functional role of prioritised photoreceptor OS genes.

## Look-up Analysis

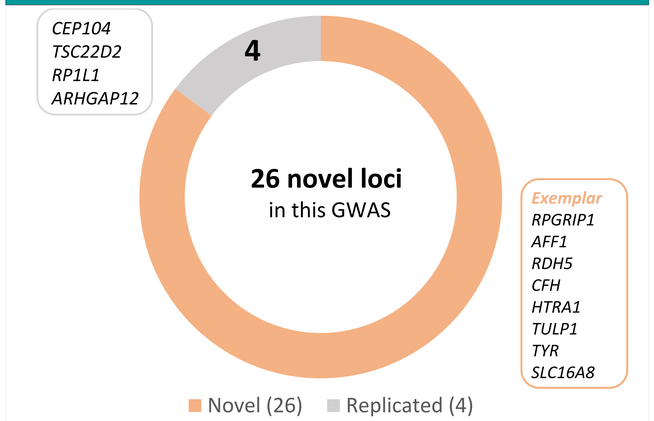



**Figure 4**: Variant lookup with respect to GWAS signals of all retina photoreceptor thickness.

## CONCLUSION

- ❑ This study identifies **30 genome-wide significant loci** for sub foveal OS thickness— **26 novel loci** and **4 replicated** from Currant et al. 2023.
- ❑ GP Map colocalisation confirms **3 loci shared with AMD**, links *RDH5* to **myopia** ($P < 10^{-60}$), and reveals pleiotropic associations with other systems including **ciliopathy**, **haematology**, and **immunity**.
- ❑ Notably, *RDH5* ($P = 9.3\times10^{-76}$) reached only $P = 5.1\times10^{-4}$ in Currant et al. 2023, demonstrating that **sub foveal-specific phenotyping amplifies foveal cone-specific signals.**
- ❑ This work is ongoing to further refine this genetic architecture and prioritise high-confidence candidate genes for downstream investigation.

## References

1. Currant, H., et al., (2023). *PLoS Genet*, *19*(2), e1010587. https://doi.org/10.1371/journal.pgen.1010587
2. Keane, P. A., et al., (2016). *PLoS One*, *11*(10), e0164095. https://doi.org/10.1371/journal.pone.0164095
3. Elmore, A. R., et al., (2026). *medRxiv*.2002.2019.26346618. https://doi.org/10.64898/2026.02.19.26346618

al574@leicester.ac.uk | www.linkedin.com/in/alvin-lirio-76a48449